

Invoice Review Advisor

An AI second set of eyes for large construction draws. Before a PM approves a \$10–20M invoice, the agent reviews it the way an excellent project manager would — against the last five invoices, the signed contract, change orders, RFIs, retainage, and project cost docs — and tells them **where the holes are**.

A \$20M invoice is too big to eyeball. Today approval relies on one person's judgment and time. The advisor gives them that judgment at scale: it reads everything a diligent PM *should* read, checks the invoice against it, and hands back a short brief of flags, context, and a recommendation. It **advises** — the human still approves.

1 WHAT THE AGENT READS BEFORE EVERY APPROVAL

- This invoice + line items
- Last 4–5 invoices on the commitment
- Signed contract + schedule of values
- Change orders (approved & pending)
- RFIs (open / disputed scope)
- Retainage terms & held-to-date
- Budget / forecast & balance to finish
- Docs: lien waivers, stored-material logs

2 WHAT IT CHECKS — THE “FIND THE HOLES” REVIEW

- 1

Over-billing: cumulative billed vs contract + approved COs — billing past the ceiling?
- 2

Progress plausibility: % billed vs actual progress & the trend of the last 5 draws.
- 3

Front-loading: early line items billed ahead of work (SOV drift across draws).
- 4

Change orders: is billed work covered by an *approved* CO — or a pending one?
- 5

RFIs: billing for scope tied to an open/unresolved RFI or disputed design.
- 6

Duplicates: same line / stored materials billed in a prior invoice.
- 7

Retainage: withheld at the correct % — and released only when due.
- 8

Rates & math: unit prices vs contract SOV; tax & period arithmetic reconcile.
- 9

Stored materials: billed but not installed — documented & on site?

3 WHAT THE PM GETS BACK

Advisory Brief · Invoice #5 — Acme Structures

\$20.4M this period · draw 5 of ~18 · generated in 40s

Contract **\$180M**

Approved COs **+\$6M**

Billed to date **\$92M (51%)**

Retainage held **\$9.2M**

HIGH

\$1.2M on line 14 (curtain wall) exceeds SOV remaining by \$300K — appears to bill **pending CO #7**, not yet approved.

HIGH

Line 9 duplicates \$450K stored materials already billed in draw #4 — possible double bill.

MED

\$800K billed against scope tied to **open RFI-113** (design unresolved).

MED

Retainage on line 22 withheld **8% vs 10%** contract — \$40K under-withheld.

NOTE

This period is **22% above** the trailing 4-draw average — verify progress.

PASSED

Math reconciles · tax correct · cumulative within contract + CO ceiling.

Recommendation: hold lines 9 & 14 (~\$1.65M) pending; the rest checks out. Each flag links to the invoice line, contract clause, or RFI it came from.

4 E-BUILDER APIS TO CONSUME → MCP TOOLS

MCP TOOL	E-BUILDER RESOURCE	
<code>get_invoice</code>	Invoices / InvoiceItems	READ
<code>list_invoice_history</code>	Invoices (by commitment)	READ
<code>get_commitment</code>	Commitments (+ SOV)	READ
<code>list_change_orders</code>	CommitmentChangeItems	READ
<code>list_rfis</code>	Process Instances (RFI type)	READ

MCP TOOL	E-BUILDER RESOURCE	
<code>get_budget_context</code>	Budgets / Forecast / Cashflow	READ
<code>search_documents</code>	Documents (contract, waivers)	READ
<code>get_process_status</code>	Process Instances	READ
<code>post_review_note</code>	Process comment / attach brief	WRITE
<code>return_or_approve*</code>	Process action (*human-led)	WRITE

Nearly all tools are **reads** — the advisor's power is assembling the picture, not writing. The only writes post the brief back into the e-Builder process and, optionally, return/approve on a human's instruction. Plus off-platform: **extract_invoice** (doc-AI for PDF line items) and **notify**.

5 WHAT THE AGENT DOES VS. ASKS

- DOES AUTOMATICALLY

 - Pulls the full evidence pack & reconciles the math.
 - Runs all nine checks; scores & ranks flags by severity + \$.
 - Writes the brief with a citation to the source for every claim.
- ASKS / ESCALATES — NEVER DECIDES

 - Any flag it raises → the PM confirms or overrides.
 - Missing rule (retainage %, tolerance) or ambiguous scope.
 - Final approval is always the human's** — the agent advises.

Next inputs to sharpen this: Are invoices PDFs needing extraction or already keyed in e-Builder? · Are RFIs & change orders in e-Builder processes? · What are the auto-flag tolerances (variance %, \$)? · Two-way or three-way (with receiving) matching? · Grounded in [REST API](#), [Cost module](#) & [entity](#) docs — confirm endpoint names against your tenant's Swagger.